

The Heaviside Method for Partial Fraction Decomposition

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1 Introduction: The Goal of Decomposition

Given a rational function $R(x) = \frac{P(x)}{Q(x)}$, where $P(x)$ and $Q(x)$ are polynomials and $\deg(P) < \deg(Q)$, the goal of partial fraction decomposition is to express $R(x)$ as a sum of simpler fractions whose denominators are based on the factors of $Q(x)$.

The Heaviside method provides a direct and efficient algorithm for determining the coefficients of this sum, often bypassing the need to solve large systems of linear equations.

2 Case 1: Non-Repeated Linear Factors

This is the simplest and most well-known application of the method.

Rule 2.1 (The "Cover-Up" Rule for Simple Roots). *Let $Q(x)$ have a simple (non-repeated) root α , so $Q(x) = (x - \alpha)Q_1(x)$ where $Q_1(\alpha) \neq 0$. The partial fraction decomposition of $R(x)$ will include a term $\frac{A}{x - \alpha}$.*

The coefficient A is found by "covering up" the $(x - \alpha)$ factor in the denominator of $R(x)$ and evaluating the remaining expression at $x = \alpha$:

$$A = \left. \frac{P(x)}{Q_1(x)} \right|_{x=\alpha} = \left. \frac{P(x)}{Q(x)/(x - \alpha)} \right|_{x=\alpha}$$

An alternative (and often faster) formulation, derived from L'Hôpital's rule, is:

$$A = \frac{P(\alpha)}{Q'(\alpha)}$$

Example 2.2. Decompose $f(x) = \frac{3x+1}{(x-1)(x+2)}$. The decomposition is $\frac{A}{x-1} + \frac{B}{x+2}$.

- **For A (root $x = 1$):**

$$A = \left. \frac{3x+1}{x+2} \right|_{x=1} = \frac{3(1)+1}{1+2} = \frac{4}{3}$$

- **For B (root $x = -2$):**

$$B = \frac{3x+1}{(x-1)} \Big|_{x=-2} = \frac{3(-2)+1}{-2-1} = \frac{-5}{-3} = \frac{5}{3}$$

Thus, $f(x) = \frac{4/3}{x-1} + \frac{5/3}{x+2}$.

3 Case 2: Repeated Linear Factors

This is where the simple "cover-up" method appears to fail for lower-order terms, and its full power, via derivatives, is revealed.

Note 3.1 (Why the Simple Method Fails). Consider the decomposition:

$$\frac{P(x)}{(x-\alpha)^k Q_1(x)} = \frac{A_1}{x-\alpha} + \frac{A_2}{(x-\alpha)^2} + \cdots + \frac{A_k}{(x-\alpha)^k} + \cdots$$

If we multiply both sides by $(x-\alpha)$ and set $x = \alpha$, all terms $A_2, \dots, A_k$ "blow up" to infinity because their denominators are not fully cancelled. **The simple method only works for the highest-power term, A_k , by multiplying by $(x-\alpha)^k$.**

The generalized Heaviside method leverages the Taylor expansion of the "covered-up" function.

Rule 3.2 (The "Derivative" Rule for Repeated Roots). *Let $Q(x)$ have a k -times repeated root α . The decomposition will include the terms:*

$$\frac{A_1}{x-\alpha} + \frac{A_2}{(x-\alpha)^2} + \cdots + \frac{A_k}{(x-\alpha)^k}$$

To find the coefficients $A_1, \dots, A_k$, define the "covered-up" function $R(x)$ by removing the entire repeated factor:

$$R(x) = \frac{P(x)}{Q(x)} (x-\alpha)^k$$

The coefficients are then given by the Taylor coefficients of $R(x)$ expanded around $x = \alpha$:

$$A_{k-m} = \frac{R^{(m)}(\alpha)}{m!} \quad \text{for } m = 0, 1, \dots, k-1$$

In reverse, this is more intuitive:

- $A_k = \frac{R^{(0)}(\alpha)}{0!} = R(\alpha)$ (This is the simple cover-up rule for the highest power)
- $A_{k-1} = \frac{R^{(1)}(\alpha)}{1!} = R'(\alpha)$
- $A_{k-2} = \frac{R^{(2)}(\alpha)}{2!} = \frac{1}{2} R''(\alpha)$
- $\dots$
- $A_1 = \frac{R^{(k-1)}(\alpha)}{(k-1)!}$

Proof (Sketch). Let $f(x) = \frac{P(x)}{Q(x)}$. Multiplying the decomposition by $(x - \alpha)^k$ gives:

$$R(x) = f(x)(x - \alpha)^k = A_1(x - \alpha)^{k-1} + A_2(x - \alpha)^{k-2} + \cdots + A_{k-1}(x - \alpha) + A_k + (\dots)$$

where the $(\dots)$ part contains all terms from other roots, which are all zero at $x = \alpha$.

We can see that this is precisely the Taylor series expansion of $R(x)$ centered at $x = \alpha$. By the definition of a Taylor series:

$$R(x) = R(\alpha) + \frac{R'(\alpha)}{1!}(x - \alpha) + \frac{R''(\alpha)}{2!}(x - \alpha)^2 + \dots$$

By matching the coefficients of the $(x - \alpha)^m$ terms, we find:

- Constant term: $A_k = R(\alpha)$
- $(x - \alpha)$ term: $A_{k-1} = R'(\alpha)$
- $(x - \alpha)^2$ term: $A_{k-2} = \frac{R''(\alpha)}{2!}$

This proves the rule $A_{k-m} = R^{(m)}(\alpha)/m!$. □

Example 3.3. Decompose $f(x) = \frac{x^2+1}{x(x-1)^2}$. The decomposition is $\frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$.

- **For A (simple root $x = 0$):** Use Rule 1.

$$A = \left. \frac{x^2+1}{-(x-1)^2} \right|_{x=0} = \frac{0^2+1}{(0-1)^2} = 1$$

- **For B, C (repeated root $x = 1$, $k = 2$):** Use Rule 2. The "covered-up" function is $R(x) = f(x)(x-1)^2 = \frac{x^2+1}{x} = x + \frac{1}{x}$.
- **For C (highest power, $m = 0$):**

$$C = A_k = A_2 = \frac{R^{(0)}(1)}{0!} = R(1) = \frac{1^2+1}{1} = 2$$

Note that since C is the parameter of the highest power of the repeated root, it can also be calculated using Rule 1, which is the simple Heaviside one. TRY YOURSELF!

- **For B (lower power, $m = 1$):** We need the first derivative: $R'(x) = \frac{d}{dx}(x + x^{-1}) = 1 - x^{-2} = 1 - \frac{1}{x^2}$.

$$B = A_{k-1} = A_1 = \frac{R^{(1)}(1)}{1!} = R'(1) = 1 - \frac{1}{1^2} = 0$$

Thus, $f(x) = \frac{1}{x} + \frac{0}{x-1} + \frac{2}{(x-1)^2} = \frac{1}{x} + \frac{2}{(x-1)^2}$.

4 Case 3: Complex and Quadratic Factors

A powerful feature of the Heaviside method is that it applies **identically** to complex roots. This provides a direct path to finding coefficients for irreducible quadratic factors, bypassing the traditional "Ax+B" coefficient matching.

Rule 4.1 (Application to Complex Roots). *The rules from Case 1 and Case 2 are valid for $x, \alpha \in \mathbb{C}$. An irreducible quadratic $x^2 + px + q$ is simply a product of two simple, conjugate complex roots $(x - \alpha)(x - \bar{\alpha})$.*

Example 4.2. Decompose $f(x) = \frac{1}{x^2+1} = \frac{1}{(x-i)(x+i)}$. The decomposition is $\frac{A}{x-i} + \frac{B}{x+i}$.

- **For A (root $x = i$):**

$$A = \frac{1}{-(x+i)} \Big|_{x=i} = \frac{1}{i+i} = \frac{1}{2i} = -\frac{i}{2}$$

- **For B (root $x = -i$):**

$$B = \frac{1}{(x-i)} \Big|_{x=-i} = \frac{1}{-i-i} = \frac{1}{-2i} = \frac{i}{2}$$

So, $\frac{1}{x^2+1} = \frac{-i/2}{x-i} + \frac{i/2}{x+i}$. We can verify this recombines to $\frac{1}{x^2+1}$.

Note 4.3 (Deriving the $Ax + B$ Form). While the complex form is often sufficient (e.g., in complex analysis), we can recover the real $Ax + B$ form by combining the complex conjugate pairs.

Given a decomposition $\frac{C}{x-\alpha} + \frac{\bar{C}}{x-\bar{\alpha}}$ (where $\bar{C}$ is the conjugate of C , a property that holds if $P(x), Q(x)$ are real polynomials):

$$\begin{aligned} \frac{C(x - \bar{\alpha}) + \bar{C}(x - \alpha)}{(x - \alpha)(x - \bar{\alpha})} &= \frac{(C + \bar{C})x - (C\bar{\alpha} + \bar{C}\alpha)}{x^2 - (\alpha + \bar{\alpha})x + \alpha\bar{\alpha}} \\ &= \frac{(2 \operatorname{Re}(C))x - (2 \operatorname{Re}(C\bar{\alpha}))}{x^2 - (2 \operatorname{Re}(\alpha))x + |\alpha|^2} \end{aligned}$$

This gives the real coefficients directly:

- $A = 2 \operatorname{Re}(C)$
- $B = -2 \operatorname{Re}(C\bar{\alpha})$

where $C = \frac{P(\alpha)}{Q'(\alpha)}$ is found using the standard Heaviside rule at *one* of the complex roots.

5 Conclusion: A Hybrid Strategy

While the derivative-based rule is universally powerful, it can be computationally expensive for high-order repeated roots. In practice, a hybrid approach is most efficient:

1. Use the **simple "cover-up" rule (Rule 1)** to find coefficients for all non-repeated factors (both real and complex).

2. Use the **simple "cover-up" rule (Rule 1)** to find the coefficient for the *highest power* of each repeated factor (e.g., A_k in Rule 2).
3. Use the **derivative rule (Rule 2)** for the lower-order terms of repeated factors (e.g., A_{k-1}, A_{k-2}).
4. If derivatives become too complex, fall back to solving the remaining unknown coefficients by **substituting one or two simple values** for x (e.g., $x = 0$ or $x = 1$) or by **matching one or two coefficients** (e.g., the leading x^n term).

Problem Set: Partial Fraction Decomposition

Instructions

Find the partial fraction decomposition for each of the following rational functions. You are **only** required to find the final decomposed form with all unknown coefficients solved.

1. Decompose:

$$F(s) = \frac{s+4}{s^2(s-2)}$$

2. Decompose:

$$F(s) = \frac{5s^2 + s + 1}{(s+1)(s^2+1)}$$

3. Decompose (Hint: Perform polynomial long division first).

$$R(x) = \frac{x^4 + 1}{x^3 - x}$$

4. Decompose:

$$R(x) = \frac{1}{(x-1)^2(x+2)(x-3)}$$

5. Decompose (Hint: Repeated irreducible quadratic factor).

$$F(s) = \frac{2s^3}{(s^2+1)^2}$$

6. Decompose:

$$R(x) = \frac{x^2 + x + 2}{x(x^2 + 2x + 5)}$$

7. Decompose (Hint: High-order repeated linear factor).

$$F(s) = \frac{s^2 + 1}{s^3(s+1)}$$

8. Decompose:

$$F(s) = \frac{1}{s^2(s^2+4)}$$

9. Decompose (Hint: $x^4 - 1 = (x^2 - 1)(x^2 + 1)$).

$$R(x) = \frac{1}{x^4 - 1}$$

10. Decompose:

$$F(s) = \frac{s+3}{(s-1)^2(s^2+s+1)}$$

Answer Key: Partial Fraction Decomposition (Detailed Solutions)

1. **Problem:** $F(s) = \frac{s+4}{s^2(s-2)}$

Decomposition Form:

$$\frac{s+4}{s^2(s-2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-2}$$

Step 1: Find C (Heaviside Cover-up for simple root $s = 2$)

$$C = \left. \frac{s+4}{s^2} \right|_{s=2} = \frac{2+4}{2^2} = \frac{6}{4} = \frac{3}{2}$$

Step 2: Find B (Heaviside Cover-up for repeated root $s = 0$, highest power)

$$B = \left. \frac{s+4}{s-2} \right|_{s=0} = \frac{0+4}{0-2} = -2$$

Step 3: Find A (Coefficient Matching) Cross-multiply the original equation:

$$s+4 = A \cdot s(s-2) + B(s-2) + C \cdot s^2$$

$$s+4 = A(s^2 - 2s) + B(s-2) + Cs^2$$

Match the coefficients for the s^2 term:

$$0 \cdot s^2 = As^2 + Cs^2 \implies 0 = A + C$$

$$A = -C \implies A = -\frac{3}{2}$$

(Alternatively, one could use the Heaviside derivative method on $R(s) = \frac{s+4}{s-2}$ to find $A = R'(0)$)

Decomposition Result

$$F(s) = -\frac{3/2}{s} - \frac{2}{s^2} + \frac{3/2}{s-2}$$

2. **Problem:** $F(s) = \frac{5s^2 + s + 1}{(s+1)(s^2+1)}$

Decomposition Form:

$$\frac{5s^2 + s + 1}{(s+1)(s^2+1)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+1}$$

Step 1: Find A (Heaviside Cover-up for simple root $s = -1$)

$$A = \left. \frac{5s^2 + s + 1}{s^2 + 1} \right|_{s=-1} = \frac{5(-1)^2 + (-1) + 1}{(-1)^2 + 1} = \frac{5 - 1 + 1}{2} = \frac{5}{2}$$

Step 2: Find B and C (Coefficient Matching) Cross-multiply the original equation:

$$5s^2 + s + 1 = A(s^2 + 1) + (Bs + C)(s + 1)$$

$$5s^2 + s + 1 = A(s^2 + 1) + B(s^2 + s) + C(s + 1)$$

Match coefficients:

- **s^2 term:** $5 = A + B \implies 5 = \frac{5}{2} + B \implies B = \frac{5}{2}$
- **Constant term (s^0):** $1 = A + C \implies 1 = \frac{5}{2} + C \implies C = -\frac{3}{2}$

Decomposition Result

$$F(s) = \frac{5/2}{s+1} + \frac{\frac{5}{2}s - \frac{3}{2}}{s^2+1}$$

3. **Problem:** $R(x) = \frac{x^4 + 1}{x^3 - x}$

Step 1: Polynomial Long Division Since $\deg(P) \geq \deg(Q)$, we must divide first. $(x^4 + 1) \div (x^3 - x) = x$ with a remainder of $x^2 + 1$.

$$\frac{x^4 + 1}{x^3 - x} = x + \frac{x^2 + 1}{x^3 - x}$$

Step 2: Decompose the Remainder

$$\frac{x^2 + 1}{x^3 - x} = \frac{x^2 + 1}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$$

Step 3: Find A, B, C (Heaviside Cover-up)

- **Find A (root $x = 0$):** $A = \left. \frac{x^2+1}{(x-1)(x+1)} \right|_{x=0} = \frac{1}{(-1)(1)} = -1$
- **Find B (root $x = 1$):** $B = \left. \frac{x^2+1}{x(x+1)} \right|_{x=1} = \frac{1^2+1}{1(1+1)} = \frac{2}{2} = 1$
- **Find C (root $x = -1$):** $C = \left. \frac{x^2+1}{x(x-1)} \right|_{x=-1} = \frac{(-1)^2+1}{-1(-1-1)} = \frac{2}{2} = 1$

Step 4: Combine Results

Decomposition Result

$$R(x) = x - \frac{1}{x} + \frac{1}{x-1} + \frac{1}{x+1}$$

4. **Problem:** $R(x) = \frac{1}{(x-1)^2(x+2)(x-3)}$

Decomposition Form:

$$\frac{1}{(x-1)^2(x+2)(x-3)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2} + \frac{D}{x-3}$$

Step 1: Find B, C, D (Heaviside Cover-up)

- **Find B (root $x = 1$):** $B = \frac{1}{(x+2)(x-3)} \Big|_{x=1} = \frac{1}{(1+2)(1-3)} = \frac{1}{(3)(-2)} = -\frac{1}{6}$
- **Find C (root $x = -2$):** $C = \frac{1}{(x-1)^2(x-3)} \Big|_{x=-2} = \frac{1}{(-2-1)^2(-2-3)} = \frac{1}{(-3)^2(-5)} = -\frac{1}{45}$
- **Find D (root $x = 3$):** $D = \frac{1}{(x-1)^2(x+2)} \Big|_{x=3} = \frac{1}{(3-1)^2(3+2)} = \frac{1}{(2)^2(5)} = \frac{1}{20}$

Step 2: Find A (Substitute a simple value, e.g., $x = 0$)

$$\begin{aligned} \frac{1}{(x-1)^2(x+2)(x-3)} \Big|_{x=0} &= \frac{1}{(-1)^2(2)(-3)} = -\frac{1}{6} \\ \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2} + \frac{D}{x-3} \Big|_{x=0} &= \frac{A}{-1} + \frac{B}{1} + \frac{C}{2} + \frac{D}{-3} \\ -\frac{1}{6} &= -A + B + \frac{C}{2} - \frac{D}{3} \\ -\frac{1}{6} &= -A - \frac{1}{6} + \frac{-1/45}{2} - \frac{1/20}{3} \\ 0 &= -A - \frac{1}{90} - \frac{1}{60} \implies A = -\left(\frac{2}{180} + \frac{3}{180}\right) = -\frac{5}{180} = -\frac{1}{36} \end{aligned}$$

Decomposition Result

$$R(x) = -\frac{1/36}{x-1} - \frac{1/6}{(x-1)^2} - \frac{1/45}{x+2} + \frac{1/20}{x-3}$$

5. **Problem:** $F(s) = \frac{2s^3}{(s^2+1)^2}$

Decomposition Form:

$$\frac{2s^3}{(s^2+1)^2} = \frac{As+B}{s^2+1} + \frac{Cs+D}{(s^2+1)^2}$$

Step 1: Cross-multiply This is the most direct method here.

$$\begin{aligned} 2s^3 &= (As+B)(s^2+1) + (Cs+D) \\ 2s^3 &= As^3 + As + Bs^2 + B + Cs + D \\ 2s^3 + 0s^2 + 0s + 0 &= As^3 + Bs^2 + (A+C)s + (B+D) \end{aligned}$$

Step 2: Match Coefficients

- **s^3 term:** $A = 2$
- **s^2 term:** $B = 0$
- **s term:** $A + C = 0 \implies 2 + C = 0 \implies C = -2$
- **Constant term:** $B + D = 0 \implies 0 + D = 0 \implies D = 0$

Decomposition Result

$$F(s) = \frac{2s}{s^2 + 1} - \frac{2s}{(s^2 + 1)^2}$$

6. **Problem:** $R(x) = \frac{x^2 + x + 2}{x(x^2 + 2x + 5)}$

Decomposition Form:

$$\frac{x^2 + x + 2}{x(x^2 + 2x + 5)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 2x + 5}$$

(Note: $x^2 + 2x + 5$ is irreducible as $b^2 - 4ac = 4 - 20 < 0$)

Step 1: Find A (Heaviside Cover-up for $x = 0$)

$$A = \left. \frac{x^2 + x + 2}{x^2 + 2x + 5} \right|_{x=0} = \frac{2}{5}$$

Step 2: Find B and C (Coefficient Matching)

$$x^2 + x + 2 = A(x^2 + 2x + 5) + (Bx + C)x$$

$$x^2 + x + 2 = A(x^2 + 2x + 5) + Bx^2 + Cx$$

Match coefficients:

- **x^2 term:** $1 = A + B \implies 1 = \frac{2}{5} + B \implies B = \frac{3}{5}$
- **x term:** $1 = 2A + C \implies 1 = 2\left(\frac{2}{5}\right) + C \implies 1 = \frac{4}{5} + C \implies C = \frac{1}{5}$

Decomposition Result

$$R(x) = \frac{2/5}{x} + \frac{\frac{3}{5}x + \frac{1}{5}}{x^2 + 2x + 5}$$

7. **Problem:** $F(s) = \frac{s^2 + 1}{s^3(s + 1)}$

Decomposition Form:

$$\frac{s^2 + 1}{s^3(s + 1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{D}{s + 1}$$

Step 1: Find D (Heaviside Cover-up for $s = -1$)

$$D = \left. \frac{s^2 + 1}{s^3} \right|_{s=-1} = \frac{(-1)^2 + 1}{(-1)^3} = \frac{2}{-1} = -2$$

Step 2: Find A, B, C (Heaviside Derivative Method for $s = 0$) Define the covered-up function $R(s) = \frac{s^2 + 1}{s + 1}$.

- **Find C (highest power, $m = 0$):** $C = \frac{R^{(0)}(0)}{0!} = R(0) = \frac{0+1}{0+1} = 1$
- **Find B (next power, $m = 1$):** $R'(s) = \frac{(2s)(s+1)-(s^2+1)(1)}{(s+1)^2} = \frac{2s^2+2s-s^2-1}{(s+1)^2} = \frac{s^2+2s-1}{(s+1)^2}$
 $B = \frac{R^{(1)}(0)}{1!} = R'(0) = \frac{-1}{1^2} = -1$
- **Find A (lowest power, $m = 2$):** $R''(s) = \frac{(2s+2)(s+1)^2-(s^2+2s-1)(2)(s+1)}{(s+1)^4} = \frac{(2s+2)(s+1)-2(s^2+2s-1)}{(s+1)^3} = \frac{2s^2+4s+2-2s^2-4s+2}{(s+1)^3} = \frac{4}{(s+1)^3}$
 $A = \frac{R^{(2)}(0)}{2!} = \frac{4/1^3}{2} = \frac{4}{2} = 2$

Decomposition Result

$$F(s) = \frac{2}{s} - \frac{1}{s^2} + \frac{1}{s^3} - \frac{2}{s+1}$$

8. **Problem:** $F(s) = \frac{1}{s^2(s^2+4)}$

Decomposition Form:

$$\frac{1}{s^2(s^2+4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{Cs+D}{s^2+4}$$

Step 1: Use substitution $u = s^2$ (Shortcut Method) This is the fastest way for this specific form.

$$\frac{1}{u(u+4)} = \frac{K_1}{u} + \frac{K_2}{u+4}$$

$$K_1 = 1/(u+4)|_{u=0} = 1/4. \quad K_2 = 1/u|_{u=-4} = -1/4. \quad \text{So, } \frac{1/4}{u} - \frac{1/4}{u+4}.$$

Step 2: Substitute back $u = s^2$

$$\frac{1/4}{s^2} - \frac{1/4}{s^2+4}$$

Comparing with the original form, we see $A = 0$, $B = 1/4$, $C = 0$, $D = -1/4$.

Decomposition Result

$$F(s) = \frac{1/4}{s^2} - \frac{1/4}{s^2+4}$$

9. **Problem:** $R(x) = \frac{1}{x^4-1}$

Decomposition Form:

$$\frac{1}{(x-1)(x+1)(x^2+1)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+1}$$

Step 1: Find A and B (Heaviside Cover-up)

- **Find A (root $x = 1$):** $A = \frac{1}{(x+1)(x^2+1)} \Big|_{x=1} = \frac{1}{(2)(2)} = \frac{1}{4}$
- **Find B (root $x = -1$):** $B = \frac{1}{(x-1)(x^2+1)} \Big|_{x=-1} = \frac{1}{(-2)(2)} = -\frac{1}{4}$

Step 2: Find C and D (Coefficient Matching)

$$1 = A(x+1)(x^2+1) + B(x-1)(x^2+1) + (Cx+D)(x^2-1)$$

Match coefficients:

- **x^3 term:** $0 = A + B + C \implies 0 = \frac{1}{4} - \frac{1}{4} + C \implies C = 0$
- **Constant term:** $1 = A(1)(1) + B(-1)(1) + D(-1) = A - B - D$

$$1 = \frac{1}{4} - \left(-\frac{1}{4}\right) - D \implies 1 = \frac{1}{2} - D \implies D = -\frac{1}{2}$$

Decomposition Result

$$R(x) = \frac{1/4}{x-1} - \frac{1/4}{x+1} - \frac{1/2}{x^2+1}$$

10. **Problem:** $F(s) = \frac{s+3}{(s-1)^2(s^2+s+1)}$

Decomposition Form:

$$\frac{s+3}{(s-1)^2(s^2+s+1)} = \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{Cs+D}{s^2+s+1}$$

Step 1: Find B (Heaviside Cover-up for $s = 1$)

$$B = \frac{s+3}{s^2+s+1} \Big|_{s=1} = \frac{1+3}{1+1+1} = \frac{4}{3}$$

Step 2: Find A (Heaviside Derivative for $s = 1$) Define $R(s) = \frac{s+3}{s^2+s+1}$.

$$R'(s) = \frac{(1)(s^2+s+1) - (s+3)(2s+1)}{(s^2+s+1)^2} = \frac{s^2+s+1 - (2s^2+7s+3)}{(s^2+s+1)^2}$$

$$R'(s) = \frac{-s^2 - 6s - 2}{(s^2+s+1)^2}$$

$$A = R'(1) = \frac{-1 - 6 - 2}{(1+1+1)^2} = \frac{-9}{3^2} = -1$$

Step 3: Find C and D (Coefficient Matching)

$$s+3 = A(s-1)(s^2+s+1) + B(s^2+s+1) + (Cs+D)(s-1)^2$$

$$s+3 = A(s^3-1) + B(s^2+s+1) + (Cs+D)(s^2-2s+1)$$

Match coefficients:

- **s^3 term:** $0 = A + C \implies 0 = -1 + C \implies C = 1$
- **Constant term:** $3 = A(-1) + B(1) + D(1) = -A + B + D$

$$3 = -(-1) + \frac{4}{3} + D \implies 3 = 1 + \frac{4}{3} + D \implies 3 = \frac{7}{3} + D$$

$$D = 3 - \frac{7}{3} = \frac{9-7}{3} = \frac{2}{3}$$

Decomposition Result

$$F(s) = -\frac{1}{s-1} + \frac{4/3}{(s-1)^2} + \frac{s+2/3}{s^2+s+1}$$